Chapter 11

Robot Control

A robot arm can exhibit a number of different behaviors, depending on the task and its environment. It can act as a source of programmed motions for tasks such as moving an object from one place to another or tracing a trajectory for a spray paint gun. It can act as a source of forces, as when applying a polishing wheel to a workpiece. In tasks such as writing on a chalkboard, it must control forces in some directions (the force must press the chalk against the board) and motions in others (the motion must be in the plane of the board). When the purpose of the robot is to act as a haptic display, rendering a virtual environment, we may want it to act like a spring, damper, or mass, yielding in response to forces applied to it.

(力是因，运动是果)

不一方向不能独立控制运动、力

4种 control strategies

① 纯运动

② 纯力

③ 混合 (运动、力方向不同)

④ 阻抗 control (不拒绝外力，而顺应产生相应动作)

Controller：将 task → actuators 的力和力矩

In each of these cases, it is the job of the robot controller to convert the task specification to forces and torques at the actuators. Control strategies that achieve the behaviors described above are known as **motion control**, **force control**, **hybrid motion–force control**, or **impedance control**. Which of these behaviors is appropriate depends on both the task and the environment. For example, a force-control goal makes sense when the end-effector is in contact with something but not when it is moving in free space. We also have a fundamental constraint imposed by the mechanics, irrespective of the environment: the robot cannot independently control the motion and force in the same direction. If the robot imposes a motion then the environment will determine the force, and if the robot imposes a force then the environment will determine the motion.

min (轨迹跟踪误差)

min (力误差)

Once we have chosen a control goal consistent with the task and environment, we can use feedback control to achieve it. Feedback control uses position, velocity, and force sensors to measure the actual behavior of the robot, com-

反馈控制

位置、速度、力 传感器

pares it with the desired behavior, and modulates the control signals sent to the actuators. Feedback is used in nearly all robot systems.

In this chapter we focus on: feedback control for motion control, both in the joint space and in the task space; force control; hybrid motion–force control; and impedance control.

11.1 Control System Overview

编码器

A typical control block diagram is shown in Figure 11.1(a). The sensors are typically: potentiometers, encoders, or resolvers for joint position and angle sensing; tachometers for joint velocity sensing; joint force–torque sensors; and/or multi-axis force–torque sensors at the "wrist" between the end of the arm and the end-effector. The controller samples the sensors and updates its control signals to the actuators at a rate of hundreds to a few thousands of Hz. In most robotic applications control update rates higher than this are of limited benefit, given the time constants associated with the dynamics of the robot and environment. In our analysis we will ignore the fact that the sampling time is nonzero and treat controllers as if they were implemented in continuous time.

3种传感器

测速电机

频率

连续时间假设

速度获取

直接测：测速电机

位置差分：$v_k = (q_k - q_{k-1})/\Delta t$

编码器

While tachometers can be used for direct velocity sensing, a common approach is to use a digital filter to numerically-difference the position signals at successive timesteps. A low-pass filter is often used in combination with the differencing filter to reduce the high-frequency signal content due to quantization of the differenced position signals.

低通滤波器

差分滤波器

滤掉差分信号高频毛刺

As discussed in Section 8.9, there are a number of different technologies for creating mechanical power, transforming the speeds and forces, and transmitting to the robot joints. In this chapter we lump each joint's amplifier, actuator, and transmission together and treat them as a transformer from low-power control signals to forces and torques. This assumption, along with the assumption of perfect sensors, allows us to simplify the block diagram of Figure 11.1(a) to the one shown in Figure 11.1(b), where the controller produces the forces and torques directly. The rest of this chapter deals with the control algorithms that go inside the "controller" box in Figure 11.1(b).

假设 controller 直接输出力/力矩

Real robot systems are subject to flexibility and vibrations in the joints and links, backlash at the gears and transmissions, actuator saturation limits, and limited resolution of the sensors. These raise significant issues in design and control but they are beyond the scope of this chapter.

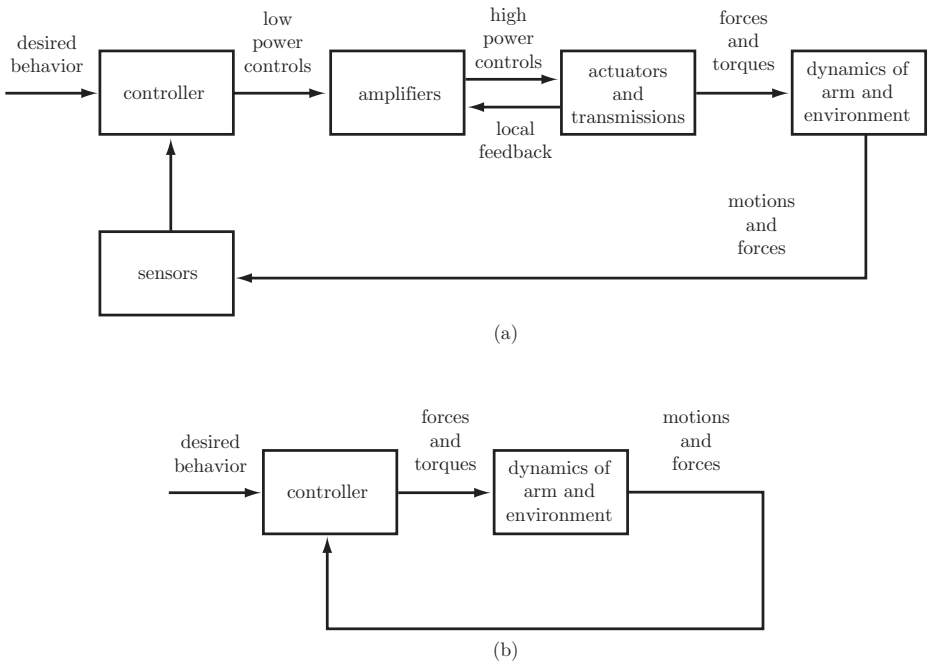

Figure 11.1: (a) A typical robot control system. An inner control loop is used to help the amplifier and actuator to achieve the desired force or torque. For example, a DC motor amplifier in torque control mode may sense the current actually flowing through the motor and implement a local controller to better match the desired current, since the current is proportional to the torque produced by the motor. Alternatively the motor controller may directly sense the torque by using a strain gauge on the motor's output gearing, and close a local torque-control loop using that feedback. (b) A simplified model with ideal sensors and a controller block that directly produces forces and torques. This assumes ideal behavior of the amplifier and actuator blocks in part (a). Not shown are the disturbance forces that can be injected before the dynamics block, or disturbance forces or motions injected after the dynamics block.

11.2 Error Dynamics 误差动力学

In this section we focus on the controlled dynamics of a single joint, as the concepts generalize easily to the case of a multi-joint robot.

If the desired joint position is $\theta_d(t)$ and the actual joint position is $\theta(t)$ then we define the joint error to be

期望 实际

误差 $$\theta_e(t) = \theta_d(t) - \theta(t).$$